



 **Time:** 6 minutes

 **Outcome:** Calculate the real exchange rate and use PPP as a long-run benchmark.

Difficulty: ●●○ Intermediate

Real Exchange Rates and Purchasing Power



THE CORE IDEA

Let e be foreign currency per domestic currency, P the domestic basket price, and P^* the price of the same basket abroad. The real exchange rate is $\epsilon = e \times P / P^*$. Higher ϵ means domestic goods are relatively more expensive: real appreciation. Purchasing-power parity (PPP) implies $\epsilon = 1$ for identical baskets under frictionless arbitrage. Transport costs, trade barriers, and nontraded goods limit this benchmark.



HOW TO RECOGNIZE IT

- Nominal e compares currencies; real ϵ compares converted basket prices.
- Higher e or P raises ϵ ; higher P^* lowers ϵ , other things equal.
- Real appreciation tends to reduce exports and raise imports.



WATCH OUT

State the quotation and compare equivalent baskets. Price indexes with arbitrary base years need not yield $\epsilon = 1$ even when comparable goods obey purchasing-power parity.



WORKED EXAMPLE: CONVERTING TWO BASKETS

A basket costs \$50 at home and 40 foreign units abroad; $e = 0.80$ foreign units per dollar. Then $\epsilon = 0.80 \times 50 / 40 = 1.00$. With e and P^* fixed, a rise in P to \$55 gives $\epsilon = 1.10$, a 10% real appreciation. Alternatively, a fall in P to \$45 gives $\epsilon = 0.90$, a 10% real depreciation.



CHECK YOURSELF

With e and P fixed, foreign basket prices rise. Does ϵ rise or fall, and is this a real appreciation or depreciation?



READY?

Return to the game and master this concept.